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SENSOR PACKAGES WITH WAVELENGTH-SPECIFIC LIGHT FILTERS

Abstract

In examples, a method comprises covering at least part of a semiconductor die having a light sensor with a mold compound, the mold compound including a cavity over the light sensor. The method further comprises depositing a solution into the cavity to form a light filter, and curing the solution to form a light filter in the cavity.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This nonprovisional application is a division of U.S. patent application Ser. No. 17/161,581, filed Jan. 28, 2021, which is hereby incorporated by reference in its entirety.

BACKGROUND
[0002] Electrical circuits are formed on semiconductor dies and subsequently packaged inside mold compounds to protect the circuits from damage due to elements external to the package, such as moisture, heat, and blunt force. To facilitate communication with electronics external to the package, an electrical circuit within the package is electrically coupled to conductive terminals. These conductive terminals are positioned inside the package but are exposed to one or more external surfaces of the package. By coupling the conductive terminals to electronics external to the package, a pathway is formed to exchange electrical signals between the electrical circuit within the package and the electronics external to the package via the conductive terminals.
SUMMARY
[0003] In examples, a method comprises covering at least part of a semiconductor die having a light sensor with a mold compound, the mold compound including a cavity over the light sensor. The method further comprises depositing a solution into the cavity to form a light filter; and curing the solution to form a light filter in the cavity.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0004] FIGS. **1**A, **1**B and **1**C are profile cross-sectional, top-down, and perspective views, respectively, of a sensor package having a wavelength-specific light filter, in accordance with various examples.
[0005] FIGS. **2**A, **2**B and **2**C are profile cross-sectional, top-down, and perspective views, respectively, of a sensor package having a wavelength-specific light filter, in accordance with various examples.
[0006] FIGS. **3**A, **3**B and **3**C are profile cross-sectional, top-down, and perspective views, respectively, of a sensor package having a wavelength-specific light filter, in accordance with various examples.
[0007] FIGS. **4**A-**4**F are a process flow for manufacturing a sensor package having a wavelength-specific light filter, in accordance with various examples.
[0008] FIG. **5** is a flow diagram of a method for manufacturing a sensor package having a wavelength-specific light filter, in accordance with various examples.
[0009] FIG. **6** is a block diagram of an electronic device including a sensor package having a wavelength-specific light filter, in accordance with various examples.
DETAILED DESCRIPTION
[0010] Some packages include sensors that are positioned on or in the semiconductor die. These sensors are configured to sense various properties in the environment, such as light, humidity, temperature, sound, potential of hydrogen (pH), and others. To sense many such properties, the sensor may be exposed to the environment, with the remainder of the package covered by an opaque mold compound. For example, to sense the pH of a liquid, the sensor may come into direct contact with the liquid. However, light sensors are not directly exposed to the environment. Instead, a transparent or translucent material may cover light sensors. This transparent or translucent material permits light to reach the sensor but simultaneously protects the sensor from damage.
[0011] In some light sensor applications, it is useful to filter (reject) specific wavelengths of light

so such light does not reach the light sensor. For example, in some cases, a transparent mold compound containing a wavelength-specific light rejection dye is used in lieu of an opaque mold compound to cover the semiconductor die, the light sensor, and other package components. While somewhat effective in filtering specific wavelengths of light, such transparent mold compounds are expensive because they entail covering all package components, rather than just the light sensor, with the transparent mold compound. In other cases, a light filter film is used to cover the light sensor. Like the transparent mold compound, this light filter film is configured to reject specific wavelengths of light. Unlike the transparent mold compound, the light filter film does not cover all package components. However, the light filter film is still expensive because of the manner in which it is applied. Specifically, the film is applied in liquid form to a wafer at the wafer stage of the package manufacturing process. The wafer is then spun to produce an even film coating. Portions of the film not covering the light sensor are then removed. This waste results in an inefficient and expensive application process.

[0012] This description presents various examples of a sensor package with a wavelength-specific light filter. The sensor package includes an opaque mold compound that covers various components of the sensor package, such as the semiconductor die. The mold compound includes a cavity, and a light sensor formed on or in the semiconductor die is exposed to this cavity. The cavity includes a transparent or translucent light filter covering the light sensor, abutting the opaque mold compound, and having a thickness of at least 0.5 millimeters. The light filter includes a combination of silicone, epoxy, metal particles, and an organic dye. This combination of materials is specifically configured to reject light having a wavelength in a target wavelength range, such as the infrared light wavelength range. The specific wavelengths filtered may be controlled by adjusting the thickness of the light filter and/or the material composition of the light filter. The light filter is deposited as a solution (e.g., using a syringe) into the cavity after the mold compound has been injected and cured, and the light filter solution is then cured. Because the light filter is applied in the area of the light sensor rather than the entire semiconductor die or the entire package, costs are mitigated relative to other solutions.

[0013] FIGS. **1**A, **1**B and **1**C are profile cross-sectional, top-down, and perspective views, respectively, of a sensor package having a wavelength-specific light filter, in accordance with various examples. Specifically, FIG. **1**A is a profile cross-sectional view of an example sensor package **100**. FIG. **1**A shows the sensor package **100** as a wire bonded quad flat no-lead (QFN) style package, but other types of packages, such as dual in-line packages (DIP), packages with gullwing-style leads, etc., also fall within the scope of this description. The sensor package **100** includes a die pad **102** and conductive terminals **104**. The sensor package **100** includes a semiconductor die **106** (e.g., a silicon die) coupled to the die pad **102**. The semiconductor die **106** includes an active surface **107** on and/or in which various circuits and/or circuit components may be formed. Bond wires **108** couple to the active surface **107** and to the conductive terminals **104**.

[0014] The sensor package **100** includes a light sensor **110** on and/or in the active surface **107**. The light sensor **110** may be any suitable type of light sensor for a given application. For example, a smartphone containing the sensor package **100** may include a light sensor **110** that is suitable for capturing ambient light that is useful to other circuitry in the smartphone to adjust output settings on a display. An opaque mold compound **112** partially or fully covers the die pad **102**, the conductive terminals **104**, the semiconductor die **106**, the active surface **107**, and the bond wires **108**. The mold compound **112** includes a cavity **114** that may be partially hollow and partially filled or that may be fully filled, as described below. The dimensions of the cavity **114** may be chosen for a specific application, with wider and/or shallower cavities permitting light from a wider angle of view to reach the light sensor **110** and narrower and/or deeper cavities permitting light from a narrower angle of view to reach the light sensor **110**. The depth of the cavity **114** may be measured from a top surface **116** of the mold compound **112** to the active surface **107**, and the width of the cavity **114** may be measured as numeral **118** indicates. The light sensor **110** may be exposed to the cavity **114**. In examples, the term exposed to means that, other than the light filter **120** (described below), no other components or materials are positioned between the light sensor **110** and the cavity **114**.

[0015] The sensor package **100** includes a light filter **120**. The light filter **120** includes a combination of silicone, an epoxy base material with metal particles, and an organic dye. The specific types, quantities, and/or proportions of silicone, epoxy base, metal particles, and organic dyes used are collectively referred to herein as the material composition of the light filter **120**. This material composition may vary and may be selected to achieve specific properties of the light filter **120**, such as the physical properties described below, as well as the ability to filter specific target wavelength ranges. Examples of the silicone material include phenyl resin silicone and methyl silicone. Examples of the epoxy base material include biphenyl and bisphenol epoxy. Examples of the metal particles include aluminum oxide and titanium oxide. Examples of the organic dyes include squalilium and phthalocyanine. In examples, the light filter **120** is translucent or transparent.

[0016] In some examples, the light filter **120** fills the entire cavity **114**, and in other examples, the light filter **120** fills part, but not all, of the cavity **114**. In some examples, the light filter **120** has a minimum thickness of 0.5 millimeters. A thinner light filter **120** may not adequately filter light in the target wavelength range, although it is possible to adjust the material composition of a light filter **120** thinner than 0.5 millimeters as described below to increase filtering efficacy. A thickness of 0.5 millimeters is not possible in other solutions using light filter films because the light filter film is applied to a wafer and the wafer is subsequently spun, resulting in a light filter film that is consistently below 0.5 millimeters in thickness. Although various thicknesses may be possible in transparent mold compound solutions, these solutions are costly, as described above. In contrast to these other solutions, the light filter **120** of the sensor package **100** covers only an area in the vicinity of the light sensor **110** (e.g., sufficient to permit adequate light from a target angle of view to reach the light sensor **110**) and is deposited into the cavity **114** after the remainder of the sensor package **100** has been assembled, resulting in an inexpensive solution that can reach thicknesses of 2.0 millimeters or more. In some examples, a thickness exceeding 5.0 millimeters may preclude adequate light from reaching the light sensor **110**.

[0017] In some examples, the material composition of the light filter **120** may be adjusted to produce a light filter **120** having specific properties, some of which may be realized prior to curing, and others which may be realized post-curing. Some of these properties are now described. For example, the light filter **120** may have a viscosity ranging from 1.0 Pascal-seconds to 1.5 Pascal-seconds at 23 degrees Celsius prior to curing. A more viscous solution may produce a light filter **120** with flaws, such as uneven thicknesses in the light filter **120**. A less viscous solution may cause patterning and contamination due to the spread of ink. In some examples, the light filter **120** may have a density ranging from 1.1 grams/cm.sup.3 to 1.3 grams/cm.sup.3. In some examples, the light filter **120** may have a hardness ranging between 80 and 90 on the type D Shore hardness scale, with a hardness less than 80 possibly resulting in cosmetic scratches on the package and possibly causing internal damage to the silicon die, and with a hardness greater than 90 being difficult to realize due to the combination of materials used to form the light filter **120**. In some examples, the light filter **120** may have a flexural strength ranging from 50 to 60 Mega Pascals, with a flexural strength below this range possibly resulting in fractures in the package, and a flexural strength above this range being difficult to realize due to the combination of materials used to form the light filter **120**. In some examples, the light filter **120** may have a flexural modulus ranging from 1400 to 1500 Mega Pascals, with a flexural modulus below this range possibly resulting in internal damage to the silicon die, and a flexural modulus above this range possibly resulting in fractures to the package responsive to the package being under mechanical stress.

[0018] In some examples, the light filter **120** may have a glass transition temperature ranging from 160 to 180 degrees Celsius, with a glass transition temperature lower than this range possibly resulting in deformation under high temperatures, and a glass transition temperature higher than this range being difficult to realize due to the combination of materials used to form the light filter **120**. In some examples, if a mismatch in coefficients of thermal expansion exists between the silicon die and other package materials such as the mold compound, cracks may result in high or low temperature conditions. In some examples, the light filter **120** may have a volume resistivity ranging from 80 to 90 tera Ohm-meters, with a volume resistivity below this range possibly resulting in breakdown due to electrostatic discharge, and a volume resistivity above this range being difficult to realize due to the combination of materials used to form the light filter **120**. In some examples, the light filter **120** has a moisture permeation ranging from 3.0 to 5.0 grams/m.sup.2-days, with a moisture permeation below this range being difficult to realize due to the combination of materials used to form the light filter **120**, and a moisture permeation above this range possibly resulting in oxidation and damage to metals inside the package, such as to bond wires.

[0019] The material composition of the light filter **120** may be specifically configured to filter light having specific target wavelength ranges. For example, the light filter **120** may be configured to reject some or all infrared light. In some examples, the light filter **120** is configured to reject light in a specific target wavelength range that is a contiguous portion of the light spectrum, and in other examples, the light filter **120** is configured to reject light in a specific target wavelength range that is divided into two or more non-contiguous portions of the light spectrum.

[0020] FIG. **1**B is a top-down view of the sensor package **100**. In examples, the light sensor **110** has a rectangular shape in the horizontal plane. In examples, the cavity **114** has a rectangular shape in the horizontal plane. As FIG. **1**B shows, the light filter **120** is transparent, making the light sensor **110** and the active surface **107** visible in this top-down view. FIG. **1**C is a perspective view of the sensor package **100**.

[0021] FIGS. **2**A, **2**B and **2**C are profile cross-sectional, top-down, and perspective views, respectively, of a sensor package having a wavelength-specific light filter, in accordance with various examples. Specifically, FIG. **2**A is a profile cross-sectional view of an example sensor package **200**. The sensor package **200** may be a QFN package, although other package types are included in the scope of this description. The sensor package **200** includes multiple conductive terminals **202**. The sensor package **200** includes a semiconductor die **204** having an active surface **206** on and/or in which various circuits and/or circuit components are formed. Conductive components **208** (e.g., solder balls) couple to the active surface **206** and to the conductive terminals **202**. The semiconductor die **204** includes an inactive surface **209**. The active surface **206** faces the conductive terminals **202**, and the inactive surface **209** faces away from the conductive terminals **202**. The inactive surface **209** includes a light sensor **210**. The semiconductor die **204** includes a conductive via **212** that extends through the thickness (e.g., the body) of the semiconductor die **204** and that is coupled to the light sensor **210** and to the active surface **206** (e.g., to a processor, microcontroller, or other suitable circuitry on the active surface **206** that can route, receive, and/or process signals from the light sensor **210**). An opaque mold compound **214** covers some or all of the conductive terminals **202**, the semiconductor die **204**, and the conductive components **208**. A cavity **216** is formed in the mold compound **214**. The properties of the cavity **216** may be similar to those of the cavity **114** described above, and thus they are not repeated here. A light filter **218** fills some or all of the cavity **216**. The properties of the light filter **218** may be similar to those of the light filter **120** described above, and thus they are not repeated here. FIG. **2**B is a top-down view of the sensor package **200**, and FIG. **2**C is a perspective view of the sensor package **200**.

[0022] FIGS. **3**A, **3**B and **3**C are profile cross-sectional, top-down, and perspective views, respectively, of a sensor package having a wavelength-specific light filter, in accordance with various examples. Specifically, FIG. **3**A is a profile cross-sectional view of an example sensor package **300**. The sensor package **300** may be a QFN package, a DIP, or any other suitable type of package. The sensor package **300** includes a die pad **302** and conductive terminals **304**. The sensor package **300** includes a semiconductor die **306** coupled to the die pad **302** and having an active surface **308**. The active surface **308** includes circuits and/or circuit components formed in and/or on the active surface **308**. Bond wires **310** are coupled to the active surface **308** and to the conductive terminals **304**. The active surface **308** includes multiple light sensors **312**, **314**, **316**, and **318**. In some examples, each of the light sensors **312**, **314**, **316**, and **318** may be similar to the light sensors **110**, **210** described above. An opaque mold compound **320** covers some or all of the die pad **302**, the conductive terminals **304**, the semiconductor die **306**, and the bond wires **310**. The mold compound **320** includes cavities **322**, **324**, **326**, and **328**. Each of the light sensors **312**, **314**, **316**, and **318** is exposed to a different one of the cavities **322**, **324**, **326**, and **328**, respectively. Each of the light sensors **312**, **314**, **316**, and **318** is partially or fully filled with a different light filter **330**, **332**, **334**, and **336**, respectively.

[0023] Each of the light filters **330**, **332**, **334**, and **336** has a different material composition and/or thickness that affects the wavelengths of light that are filtered. Specifically, each of the light filters **330**, **332**, **334**, and **336** includes a different combination of silicone, epoxy base with metal particles, and organic dye and/or has a different thickness, such that each of the light filters **330**, **332**, **334**, and **336** rejects light having a different target wavelength range (such different ranges may overlap with each other). For example, the material composition and/or thickness of the light filter **330** may be set so the light filter **330** is configured to reject all light (including visible and invisible light, such as infrared light) except red light. For example, the material composition and/or thickness of the light filter **332** may be set so the light filter **332** is configured to reject all light (including visible and invisible light, such as infrared light) except green light. For example, the material composition and/or thickness of the light filter **334** may be set so the light filter **334** is configured to reject all light (including visible and invisible light, such as infrared light) except blue light. For example, the material composition and/or thickness of the light filter **336** may be set so the light filter **336** is configured to reject all light (including visible and invisible light, such as infrared light) except indigo light. By permitting only light with differing wavelengths to enter into each of the cavities **322**, **324**, **326**, and **328** and to reach the corresponding light sensors **312**, **314**, **316**, and **318**, the information obtained by each of the light sensors **312**, **314**, **316**, and **318** is more color-specific. This increased granularity in color-specific information may enable an electronic device containing the sensor package **300** to perform operations that it would otherwise be unable to perform.

[0024] In examples, the thickness of each of the light filters **330**, **332**, **334**, and **336** is at least 0.5 millimeters, as described above with reference to the light filters **120**, **218**. The cavities **322**, **324**, **326**, and **328** have properties similar to the properties of the cavity **114** as described above, and thus these properties are not described again here. Each of the light filters **330**, **332**, **334**, and **336** may have physical properties similar to the properties of the light filter **120** as described above, and thus such properties are not described again here, with the exception that the specific material compositions of the light filters **330**, **332**, **334**, and **336** may vary so different wavelengths of light are rejected. FIG. **3**B is a top-down view of the sensor package **300**, and FIG. **3**C is a perspective view of the sensor package **300**.

[0025] FIGS. **4**A-**4**F are a process flow for manufacturing a sensor package having a wavelength-specific light filter, in accordance with various examples. The process flow is presented in the context of the example sensor package **100** described above, but the process flow may be adapted to manufacture any sensor package described herein, as well as variations of the sensor packages described herein. FIG. **5** is a flow diagram of a method **500** for manufacturing a sensor package having a wavelength-specific light filter, in accordance with various examples. The method **500** is now described in parallel with the process flow of FIGS. **4**A-**4**F. The method **500** includes obtaining a die pad and a conductive terminal (**502**). FIG. **4**A is a profile cross-sectional view of the die pad **102** and multiple conductive terminals **104**. The method **500** includes coupling a semiconductor die to the die pad, the semiconductor die having a light sensor (**504**). FIG. **4**B is a profile cross-sectional view of the semiconductor die **106** having a light sensor **110** and coupled to the die pad **102**. For example, a die attach layer (not expressly shown) is useful to couple the semiconductor die **106** to the die pad **102**. The method **500** includes coupling a bond wire to the semiconductor die and to the conductive terminal (**506**). FIG. **4**C is a profile cross-sectional view of the bond wires **108** coupled to the conductive terminals **104** and to the semiconductor die **106**.

[0026] The method **500** includes positioning the die pad, the conductive terminal, the semiconductor die, and the bond wire in a mold chase (**508**). The method **500** also includes lowering a mold chase member having a dam toward the semiconductor die such that the dam covers the light sensor (**510**). FIG. **4**D is a profile cross-sectional view of the structure of FIG. **4**C positioned inside a mold chase having a mold chase bottom member **400** and a mold chase top member **402**. The mold chase top member **402** includes a dam **404**, which, responsive to the mold chase being closed, extends to and contacts the light sensor **110** at the active surface **107** of the semiconductor die **106**.

[0027] The method **500** includes injecting a mold compound into the mold chase to cover the die pad, the conductive terminal, the semiconductor die, and the bond wire(s) with the mold compound (**512**). The dam is configured to prevent the mold compound from covering the light sensor (**512**). FIG. **4**E is a profile cross-sectional view of a mold compound **112** covering some or all of the die pad **102**, the conductive terminals **104**, the semiconductor die **106**, and the bond wires **108**. The dam **404** prevented mold compound **112** from flowing over and covering the light sensor **110** and areas of the active surface **107** circumscribing the light sensor **110**. Consequently, the cavity **114** is formed. Cavity **114** has dimensions commensurate with the dimensions of the dam **404**. Thus, the dimensions of the cavity **114** may be manipulated by adjusting the dimensions of the dam **404**. The light sensor **110** is exposed to the cavity **114**.

[0028] The method **500** includes depositing a light filter solution including a combination of silicone, an epoxy base including metal particles, and an organic dye onto the light sensor (**514**). The combination is configured to reject light having a wavelength in a target wavelength range (**514**). The method **500** also includes curing the light filter solution to form a light filter (**516**). FIG. **4**F is a profile cross-sectional view of the structure of FIG. **4**E with the light filter **120** having been deposited in liquid form (e.g., by syringe deposition) in the cavity **114** and cured. In examples, the liquid light filter solution may be cured by applying heat in the range of 150-200 degrees Celsius for a time period ranging from 30-60 minutes.

[0029] FIG. **6** is a block diagram of an electronic device **600** including a sensor package having a wavelength-specific light filter, in accordance with various examples. The electronic device **600** may be a smartphone, a tablet, a laptop computer, a desktop computer, a camera, a webcam, or any other suitable device that may have use for ambient light data. The electronic device **600** includes a system board **602**. Circuitry **604** is mounted on the system board **602** and may include memory (e.g., including executable code), a processor, a microcontroller, a graphics processor, a sound card, connections (e.g., copper traces) between such components, etc. A sensor package **606** may be mounted on the system board **602** and coupled to the circuitry **604**. The sensor package **606** is representative of the example sensor packages **100**, **200**, and **300** described herein.

[0030] In this description, the term “couple” may cover connections, communications, or signal paths that enable a functional relationship consistent with this description. For example, if device A generates a signal to control device B to perform an action, then: (a) in a first example, device A is directly coupled to device B; or (b) in a second example, device A is indirectly coupled to device B through intervening component C, if intervening component C does not substantially alter the functional relationship between device A and device B, so device B is controlled by device A via the control signal generated by device A.

[0031] A device that is “configured to” perform a task or function may be configured (e.g., programmed and/or hardwired) at a time of manufacturing by a manufacturer to perform the function and/or may be configurable (or reconfigurable) by a user after manufacturing to perform the function and/or other additional or alternative functions. The configuring may be through firmware and/or software programming of the device, through a construction and/or layout of hardware components and interconnections of the device, or a combination thereof.

[0032] A circuit or device that is described herein as including certain components may instead be adapted to be coupled to those components to form the described circuitry or device. For example, a structure described as including one or more semiconductor elements (such as transistors), one or more passive elements (such as resistors, capacitors, and/or inductors), and/or one or more sources (such as voltage and/or current sources) may instead include only the semiconductor elements within a single physical device (e.g., a semiconductor die and/or integrated circuit (IC) package) and may be adapted to be coupled to at least some of the passive elements and/or the sources to form the described structure either at a time of manufacture or after a time of manufacture by an end-user and/or a third-party.

[0033] While certain components may be described herein as being of a particular process technology, these components may be exchanged for components of other process technologies. Circuits described herein are reconfigurable to include the replaced components to provide functionality at least partially similar to functionality available prior to the component replacement. Components shown as resistors, unless otherwise stated, are generally representative of any one or more elements coupled in series and/or parallel to provide an amount of impedance represented by the shown resistor. For example, a resistor or capacitor shown and described herein as a single component may instead be multiple resistors or capacitor, respectively, coupled in parallel between the same nodes. For example, a resistor or capacitor shown and described herein as a single component may instead be multiple resistors or capacitor, respectively, coupled in series between the same two nodes as the single resistor or capacitor. Unless otherwise stated, “about,” “approximately,” or “substantially” preceding a value means+/−10 percent of the stated value. Modifications are possible in the described examples, and other examples are possible within the scope of the claims.

Claims

1. A method, comprising: covering at least part of a semiconductor die having a light sensor with a mold compound, the mold compound including a cavity over the light sensor; depositing a solution into the cavity to form a light filter; and curing the solution to form a light filter in the cavity.

2. The method of claim 1, wherein covering at least part of a semiconductor die having a light sensor with a mold compound includes: positioning the semiconductor die in a mold chase; lowering a mold chase member having a dam toward the semiconductor die such that the dam covers the light sensor; and injecting the mold compound into the mold chase to cover the semiconductor die with the mold compound, in which the dam defines the cavity.

3. The method of claim 2, wherein the dam has a rectangular footprint.

4. The method of claim 1, wherein the mold compound is opaque.

5. The method of claim 1, wherein the solution includes at least one of: silicone, an epoxy including metal particles, or an organic dye.

6. The method of claim 5, wherein the solution includes a combination of the silicone, the epoxy, and the organic die, the combination configurable to reject light having a wavelength in a target wavelength range.

7. The method of claim 1, wherein curing the solution includes applying heat in the range of 150-200 degrees Celsius for a time period ranging from 30-60 minutes.

8. The method of claim 1, wherein the filter has properties including at least one of: a viscosity ranging from 1.0 Pascal-seconds to 1.5 Pascal-seconds at 23 degrees Celsius prior to curing; a hardness ranging between 80 and 90 on a type D Shore hardness scale; a flexural strength ranging from 50 to 60 Mega Pascals; a flexural modulus ranging from 1400 to 1500 Mega Pascals; a glass transition temperature ranging from 160 to 180 degrees Celsius; a volume resistivity ranging from 80 to 90 tera Ohm-meters; or a moisture permeation ranging from 3.0 to 5.0 grams/m.sup.2-days.

9. The method of claim 1, further comprising coupling the semiconductor die to a die pad, and covering at least parts of the semiconductor die and the die pad with the mold compound.

10. The method of claim 9, wherein coupling the semiconductor die to a die pad includes coupling the semiconductor die to the die pad via a die attach layer.

11. The method of claim 1, wherein depositing a solution into the cavity to form a light filter includes depositing the solution by a syringe.

12. The method of claim 1, wherein the light sensor is a first light sensor, the cavity is a first cavity, the solution is a first solution, the light filter is a first light filter, the semiconductor die has a second light sensor, the mold compound has a second cavity over the second light sensor, and the method further comprises depositing a second solution into the second cavity to form a second light filter, and curing the second solution to form a second light filter in the second cavity.

13. The method of claim 12, wherein the first and second solutions have different depths in the first and second cavities.

14. The method of claim 12, wherein the first and second solutions have different compositions.

15. The method of claim 1, further comprising forming conductive terminals and forming metal interconnects between the semiconductor die and the conductive terminals, and wherein covering at least part of a semiconductor die having a light sensor with a mold compound includes covering at least parts of the semiconductor die, the conductive terminals and the metal interconnects with the mold compound.
